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CASE STUDY

KHAVDA HYBRID PLANT

INDUSTRY	RENEWABLE ENERGY
CUSTOMER	Adani Green Energy Limited
OBJECTIVE	Plant SCADA (solar) with Hybrid PPCs & Tracker SCADA of 1300MW Hybrid Plant
LOCATION	Khavda, Gujarat
PROJECT	Khavda Hybrid Plant Project

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CHALLENGES

- Single largest project of hybrid applications (1st in India)
- Designing Plant SCADA with Hybrid PPC and Tracker SCADA system for 1300MW
- All block level events and WTG data reporting with time stamps at WRLDC Server
- Hardware-in-the-Loop tests of PPC
- Optimizing PPC loop time for plant control
- Controlling of SVGs in accordance to PPC commands





- Cyber security implementation
- WRLDC and CTU standards compliance
- Protection against environmental conditions
- Centralized plant operation from Common Control Room (CCR)
- Synchronizing plant events with WRLDC over IEC 60870-5-104
- Ethernet network with multiple ring topology for better redundancy

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SOLUTION



Redundant SCADA:
- For Tracker and Solar Plant
- Consistent data availability and accuracy
- Handle up to 8,50,000 tags
- Web enabled control and monitoring
- Cyber security compliance
- Scalable and flexible SCADA architecture

Hot Standby Controller:
- For switchyard room, main control room and PPC
- High processing power
- Highly reliable network configuration
- Multiple communication protocol supported

Optimized Loop Time:
- Enhancing the input scanning of PQM data
- PPC data processing
- Output writing to slave devices (inverters etc)

Customized PPC Logic:
- Common station-based PQM
- Draw a reference from a single point (PQM)
- SVG Integration
- Distribute commands to the slave plant's PPCs and SVGs

Hardware-in-the-Loop Testing at CPRI to validate PPC response

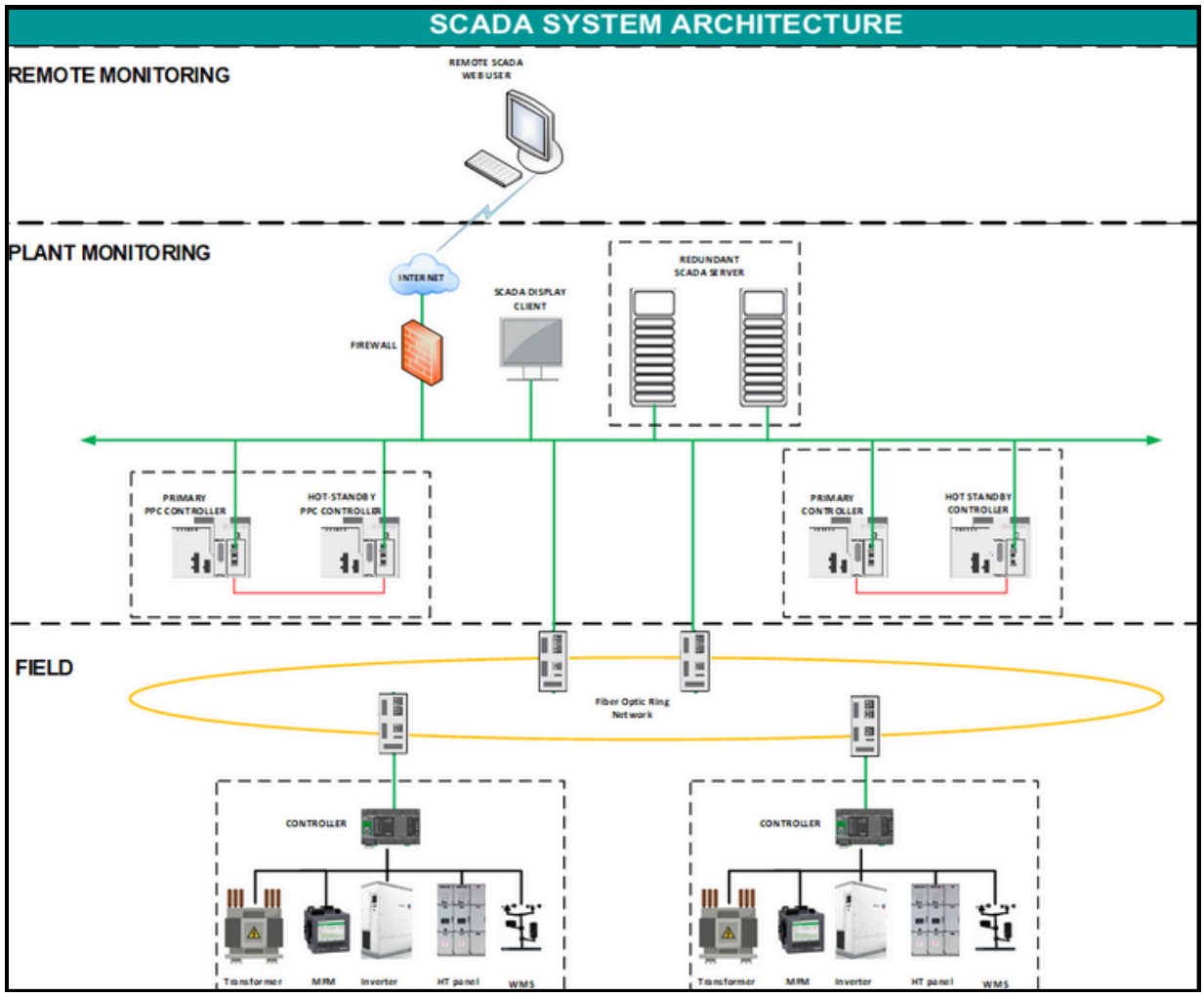
Corrosion-Resistant Design:
- Conformal coated controller and network devices
- Enclosures with corrosion-resistant paint

Redundant Gateway for WRLDC data synchronization:
- Communication with WRLDC Server over IEC 104 protocol
- All block level events & WTG data reporting with time stamps over IEC 104 protocol secure firewall policies



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ARCHITECTURE



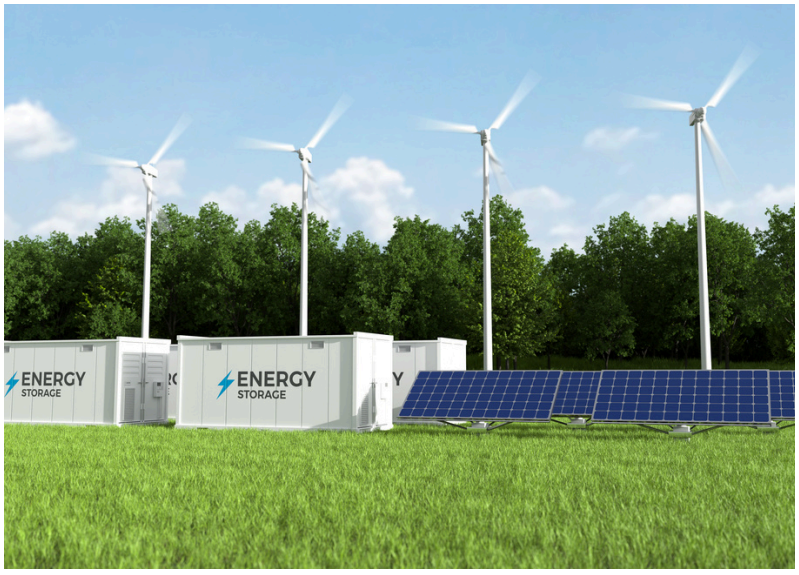


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VALUE CREATION

- **Minimized downtime**
- **Intuitive SCADA data screen**
- **Increased data integrity and availability**
- **Load balancing and improved performance**
- **Corrosion-Resistant System**
  - Reducing maintenance costs
  - Extended plant lifespan





- **Regulatory Compliance** - To obtain approvals for plant charging
- **Fault-Tolerant Network** - Ensure high reliability and availability of operations
- **Comprehensive Compliance** - With parameters viz. active power, reactive power, power factor, voltage and frequency

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